Video Wall Controller

Quick Start Guide

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Preface

Applicable Models

This manual is applicable to the DS-C66S series video wall controller.

Default Parameters

<table>
<tr><th>Type</th><th>Default Parameter</th></tr>
<tr><td>Device</td><td rowspan="2">● Login user name: admin
● IP address: 192.0.0.64</td></tr>
<tr><td>SSH connection</td></tr>
</table>

Caution

To improve system security, it is highly recommended to change password regularly. In order to protect your privacy and corporate data and avoid network security issues, it is recommended to set strong password that meets security requirements.

Symbol Conventions

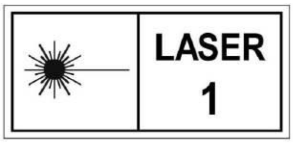
The symbols that may be found in this document are defined as follows.

Symbol	Description
Note	Provides additional information to emphasize or supplement important points of the main text.
Caution	Indicates a potentially hazardous situation, which if not avoided, could result in equipment damage, data loss, performance degradation, or unexpected results.
Danger	Indicates a hazard with a high level of risk, which if not avoided, will result in death or serious injury.

Safety Instructions

Caution

- The device must be connected to an earthed mains socket-outlet.
- The socket-outlet shall be installed near the device and shall be easily accessible.
- Do not touch the bare components (such as the metal contacts of the inlets) and wait for at least 5 minutes, since electricity may still exist after the device is powered off.
- Never place the device in an unstable location. The device may fall, causing serious personal injury or death.
- This device is not suitable for use in locations where children are likely to be present.
- CAUTION: Risk of explosion if the battery is replaced by an incorrect type.
- Improper replacement of the battery with an incorrect type may defeat a safeguard (for example, in the case of some lithium battery types).
- Do not dispose of the battery into fire or a hot oven, or mechanically crush or cut the battery, which may result in an explosion.
- Do not leave the battery in an extremely high temperature surrounding environment, which may result in an explosion or the leakage of flammable liquid or gas.
- Do not subject the battery to extremely low air pressure, which may result in an explosion or the leakage of flammable liquid or gas.
- Dispose of used batteries according to the instructions.
- Keep body parts away from fan blades. Disconnect the power source during servicing.
- Class 1 laser product used with compatible Class 1 fiber optical transceivers according to IEC 60825-1:2014 and EN 60825-1:2014+A11:2021, and hazard level 1 based on IEC 60825-2:2021 and EN 60825-2:2004+A1:2007+A2:2010. Make sure that the power has been disconnected before you wire, install, maintain, or repair. When any laser equipment is in use, make sure that the device lens is not exposed to the laser beam, or it may burn out. The laser radiation emitted from the device can cause eye injuries, burning of skin or inflammable substances. Before enabling the laser ranging function, make sure no human or inflammable substances are in front of the laser lens. Do not place the device where minors can fetch it.



Note

- This device is suitable for use in equipment room only.
- Make sure that the power has been disconnected before you wire, install, or disassemble the device.

- The device shall not be exposed to water dripping or splashing, and no objects filled with liquids, such as vases, shall be placed on the device.
- No naked flame sources, such as lighted candles, should be placed on the device.
- If smoke, odor, or noise arises from the device, immediately turn off the power, unplug the power cable, and contact the service center.
- Install the device according to the instructions in Quick Start Guide.
- To prevent injury, this device must be securely attached to the installation surface in accordance with the installation instructions.
- The ventilation should not be impeded by covering the ventilation openings with items, such as newspapers, table-cloths, curtains. The openings shall never be blocked by placing the device on a bed, sofa, rug, or other similar surface.

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Chapter 1 Introduction

1.1 Overview

The video wall controller (hereinafter referred as the device) is the core control device of the screen splicing control system. As a new-generation FPGA-based pure hardware image processing device, it adopts the structure of main control board and service boards to provide the following advantages:

- Supports the video input and video output via various ports.
- Supports the network encoding and real-time preview of signal sources.
- Supports the decoding and output of various network signal sources.
- Supports the high-definition (HD) video splicing and fusion.
- Supports the window splicing, roaming window, and other operations.
- Supports the management on users, network, operation, alarm and logs.

1.2 Appearance

1.2.1 Host System

The device adopts a plug-and-play modular design, and the host system achieves different functions by being equipped with various service boards.

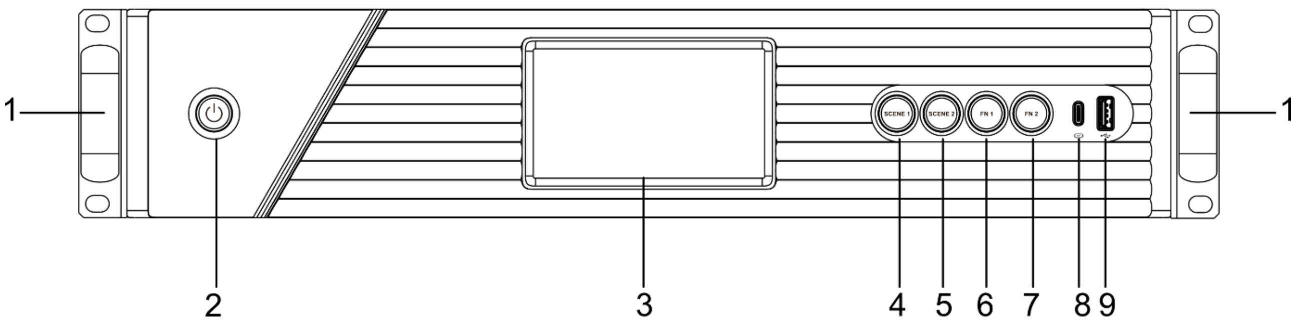

Figure 1-1 2U Device Front Panel

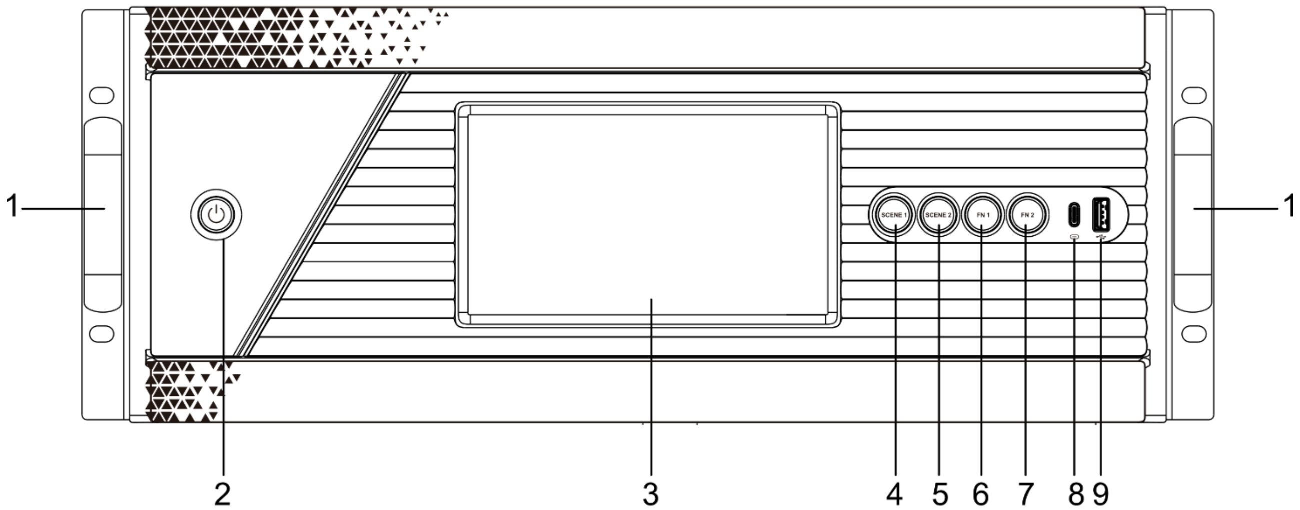

Figure 1-2 4U Device Front Panel

Table 1-1 Front Panel Description

No.	Name	Description
1	Mounting ears	Hold the handles on the mounting ears with both hands to move the device.
2	Power button/power indicator	● Power button: ● Default: The device starts automatically upon power connection (including after reconnection). ● Power off: Press and hold for 3 seconds to force shutdown (effective in any state). ● Power on: Press briefly (1 second) to power on the device (effective only when powered off). ● Power indicator status: ● On: Device is powered on. ● Off: Device is powered off.
3	LCD touch panel	● Status monitoring: Displays real-time device operation status and board status. ● Function configuration: Supports scene changing, USB upgrade, debugging information export, and quick self-test.
4	Scene switch button (SCENE 1)	One-key switch to Scene 1 of Video Wall 1.
5	Scene switch button (SCENE 2)	One-key switch to Scene 2 of Video Wall 1.
6	Custom scene switch button (FN 1)	One-key switch to the video wall scene configured

No.	Name	Description
		via the web interface.
7	Custom scene switch button (FN 2)	One-key switch to the video wall scene configured via the web interface.
8	Type-C port	Reserved.
9	USB 2.0 port	Connect a USB drive to support USB upgrade, debugging information export, and quick self-test.

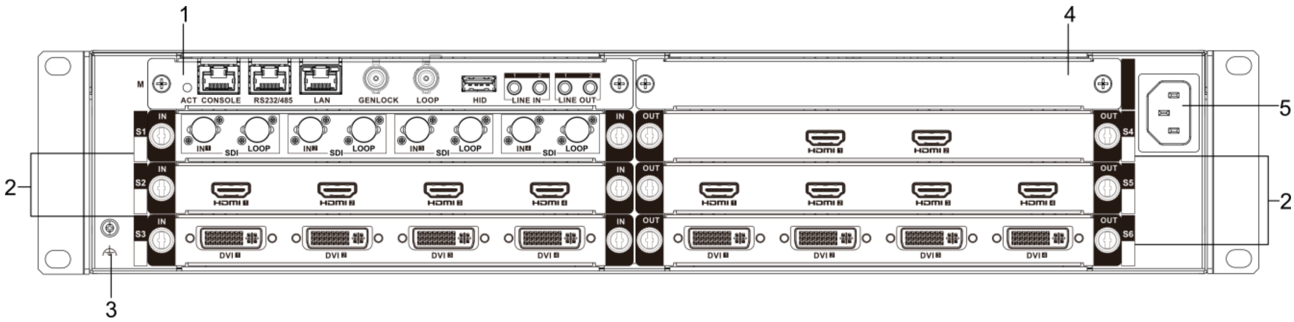

Figure 1-3 2U Device Rear Panel

Table 1-2 2U Device Rear Panel Description

No.	Name	Description
1	Main control board slot (M)	Supports the main control board.
2	Service board slot (S1 to S6)	Supports the input boards and output boards.
3	Grounding point	Connect the ground wire.
4	Empty slot	Keep the blank panel in the slot.
5	AC power input	Connects to the AC power cord.

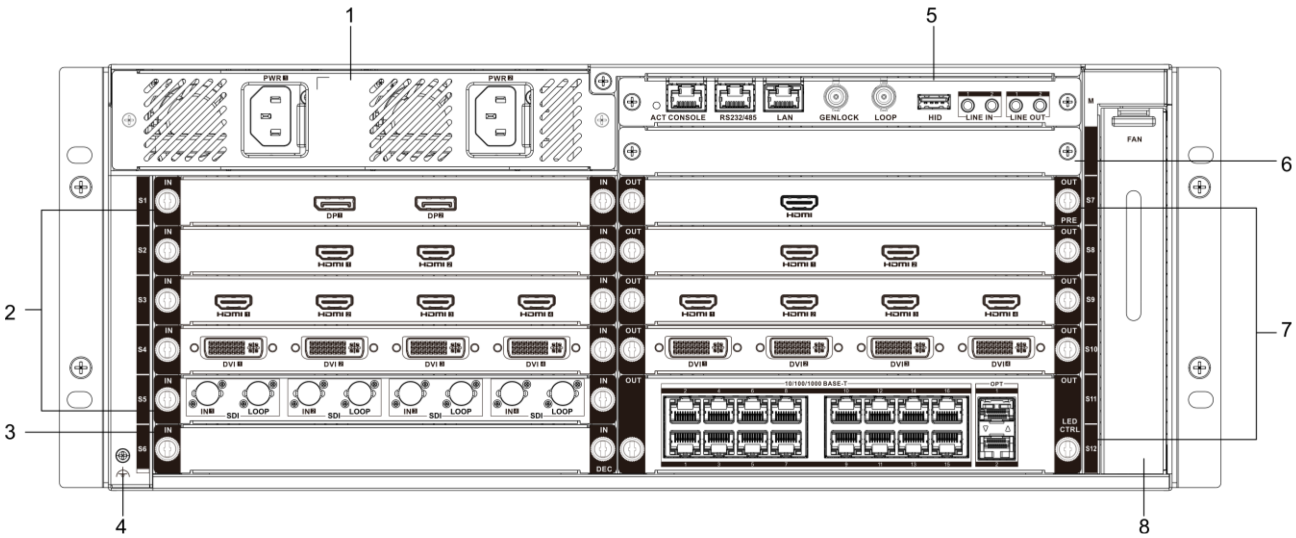

Figure 1-4 4U Device Rear Panel

Table 1-3 4U Device Rear Panel Description

No.	Name	Description
1	Power module slot (PWR1 and PWR2)	● Supports the power modules. ● Provides two power slots (one power module is provided). Supports power redundancy by adding an optional power module.
2	Service board slot (S1 to S5)	Supports the input boards.
3	Service board slot (S6)	Supports the input board, preview board, electrical LED controller board, and optical LED controller board.
4	Grounding point	Connect the ground wire.
5	Main control board slot (M)	Supports the main control board.
6	Empty slot	Keep the blank panel in the slot.
7	Service board slot (S7 to S12)	Supports the output boards.
8	Fan slot	Supports the fan module.

1.2.2 Main Control Board

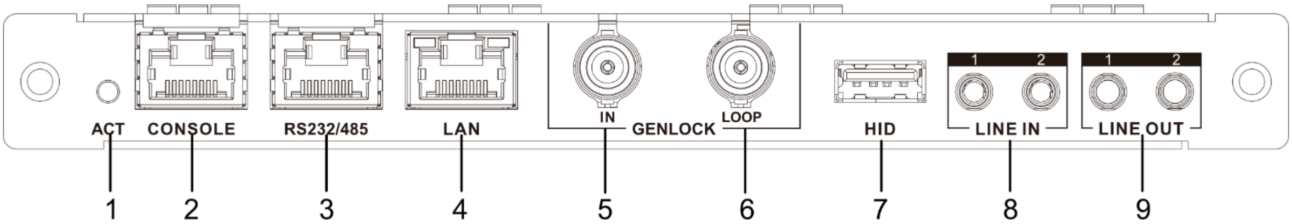

Figure 1-5 F Main Control Board

No.	Name	Description
1	ACT indicator	Flashing green: the board runs normally.
2	Console port	Connect an RJ-45 serial cable for device debugging, parameter configuration, and etc.
3	RS-485/RS-232 port	Connect an RJ-45 serial cable to an external device that supports RS-485 or RS-232 protocols.
4	Gigabit Ethernet port (LAN)	Connect a network cable.
5	Genlock input port (GENLOCK IN)	Connect to the GENLOCK port of other devices of the same type.
6	Genlock loop output port (GENLOCK LOOP)	Connect to the LOOP port of other devices of the same type for signal looping.
7	USB port (HID)	Connect to the USB port of the controlled device (such as computers, ultra-high-resolution servers, etc.) for transmitting keyboard and mouse data.
8	Audio input port (LINE IN)	Provides two audio input ports for connecting active audio sources, such as active microphone.
9	Audio output port (LINE OUT)	Provides two audio output ports for connecting to the amplified audio playback devices.

1.2.3 Input Boards

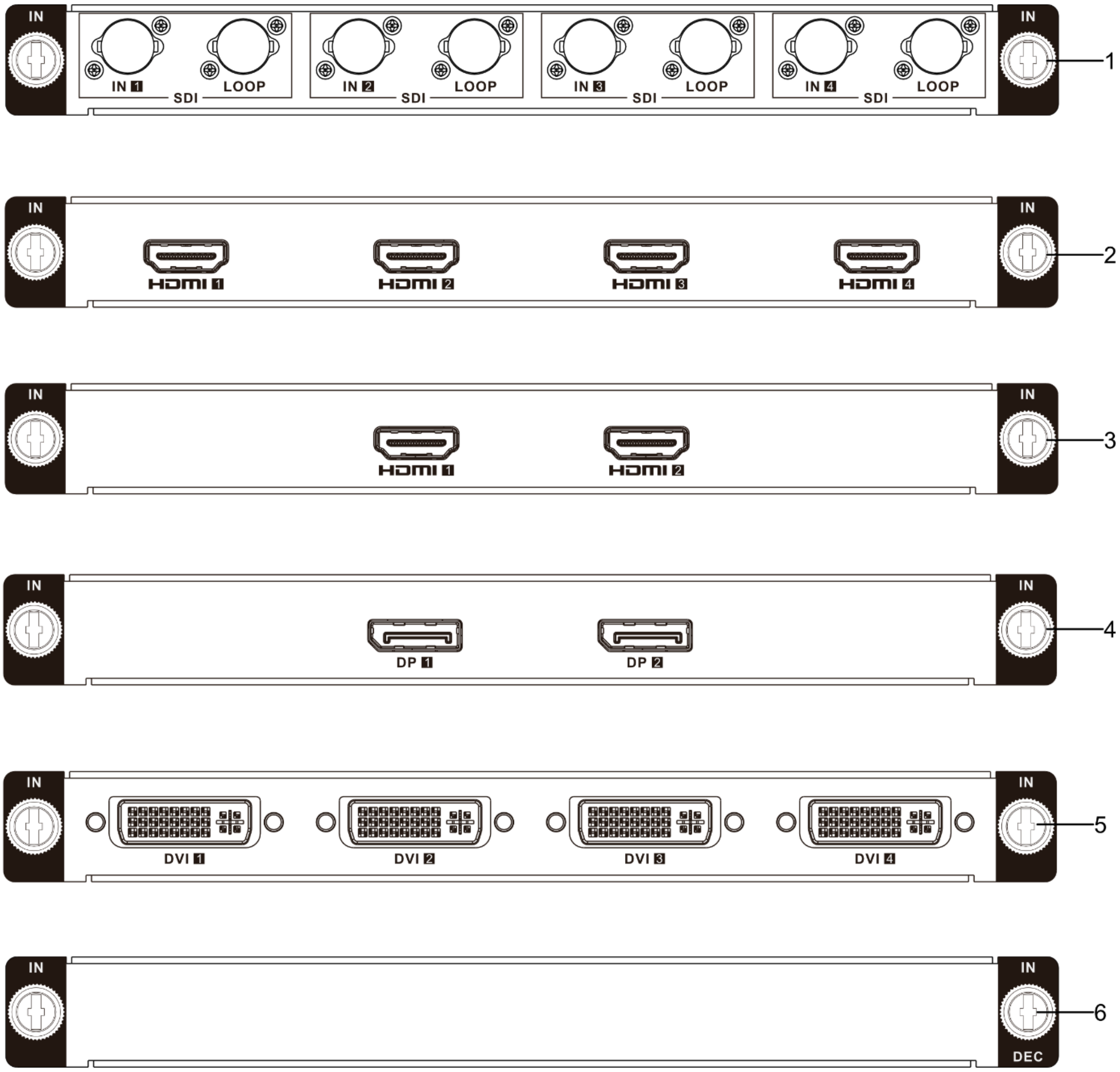

Figure 1-6 Input Boards

No.	Name	Description
1	SDI input board	● Supports 4 channels of SDI input, with a maximum input resolution of 4096 × 2160 @ 60 Hz. ● Supports 4 channels of SDI loop-through output, with a maximum output resolution of 4096 × 2160 @ 60 Hz.
2	2K HDMI input board	Supports 4 channels of HDMI input, with a maximum input resolution of 1920 × 1200 @ 60 Hz.
3	4K HDMI input board	Supports 2 channels of 4K HDMI input, with a maximum input resolution of 4096 × 2160 @ 60 Hz.
4	4K DP input board	Supports 2 channels of 4K DP input, with a maximum input

No.	Name	Description
		resolution of 4096 × 2160 @ 60 Hz.
5	DVI input board	Supports 4 channels of DVI input, with a maximum input resolution of 1920 × 1200 @ 60 Hz.
6	Decoding board	Provides 24 channels of 1080p @ 30 fps video decoding capacity.

1.2.4 Output Boards

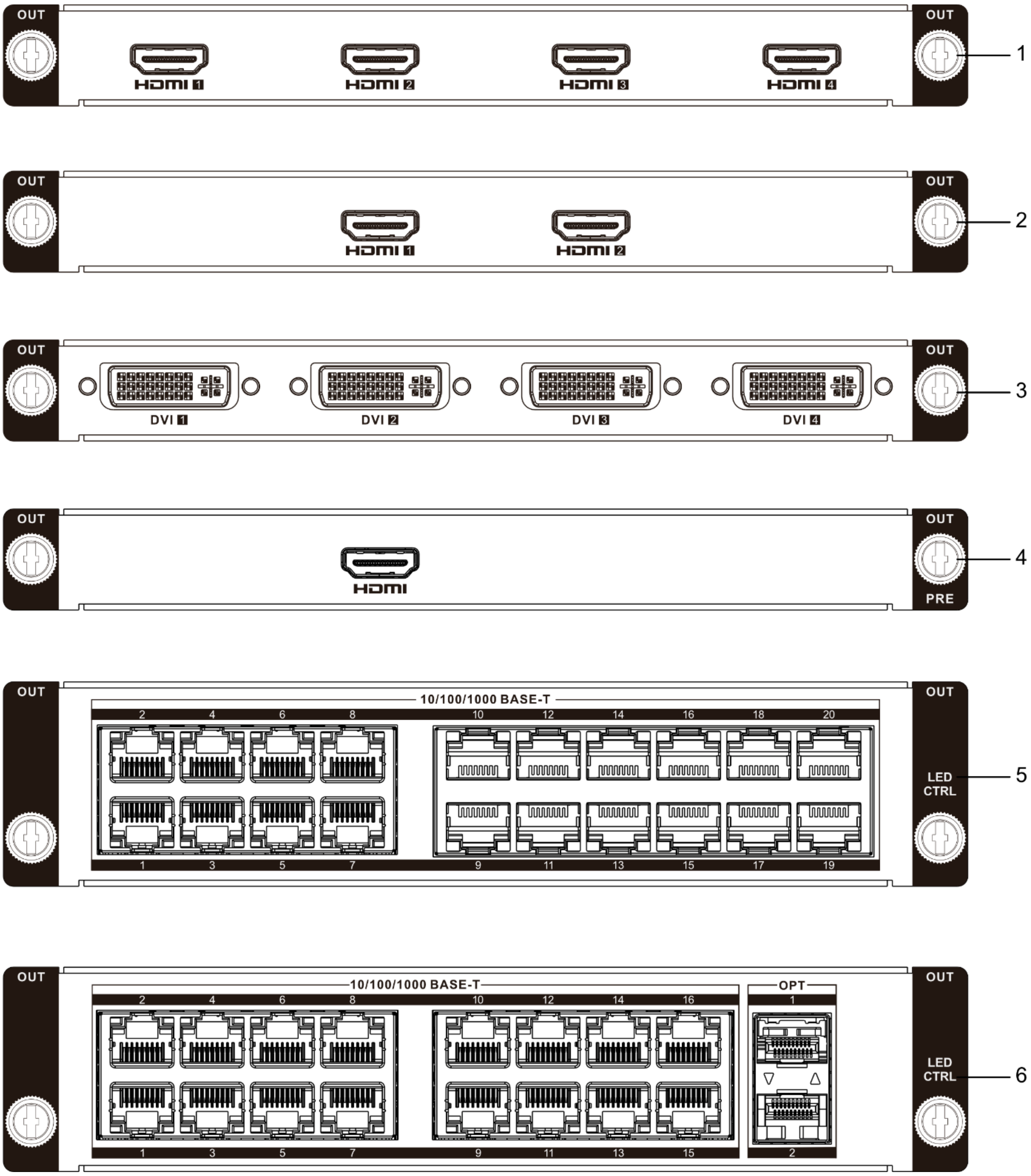

Figure 1-7 Output Boards

No.	Name	Description
1	2K HDMI output board	Provides 4 HDMI output ports for connecting HDMI display devices, with a maximum output resolution of 1920 × 1200 @ 60 Hz.
2	4K HDMI output board	Provides 2 HDMI output ports for connecting HDMI display devices, with a maximum output resolution of 4096 × 2160 @ 60 Hz.
3	DVI output board	Provides 4 DVI output ports for connecting DVI display devices, with a maximum output resolution of 1920 × 1200 @ 60 Hz.
4	Preview board	● Provides 1 HDMI output port for connecting an HDMI display device, with a maximum output resolution of 1920 × 1080 @ 60 Hz. ● Supports previewing a single video wall remotely via the client.
5	Electrical LED controller board	Provides 20 Gigabit Ethernet ports for direct connection to LED cabinets via network cables. Each port supports a maximum load of 0.65MP, and the entire board supports a total load of up to 10.4MP. **Note** This board occupies 2 slots.

6	Optical LED controller board	● Provides 16 Gigabit Ethernet ports for direct connection to LED cabinets via network cables. Each port supports a maximum load of 0.65MP, and the entire board supports a total load of up to 10.4MP. The supported image width ranges from 64 to 16,384 pixels (must be a multiple of 4), and the height ranges from 64 to 8,192 pixels.
		● Provides two 10G optical ports. Insert a 10G optical transceiver module and connect via fiber to an external LED controller.
		  ● Optical port 1 (OPT 1): Replicates data from Gigabit Ethernet ports 1 to 8.
		  ● Optical port 2 (OPT 2): Replicates data from Gigabit Ethernet ports 9 to 16.
		Note
		● This board occupies 2 slots.
		● Optical and electrical ports are mutually exclusive and cannot be used simultaneously.

Chapter 2 Installation

2.1 Safety Precautions

Caution

As a high-precision, system-level electronic product, the device should be installed and maintained by professionals.

In order to avoid personal and property injury, please read the safety precautions in this section carefully before installation. The following safety recommendations do not cover all possible dangerous situations.

Electricity Safety

- During the installation, wiring, disassembly, and maintenance of the device, please disconnect the power supply and do not operate with electricity (except for the operation of the hot plug).
- In the installation and use of the device, make sure to follow the local electrical safety regulations.
- In case of abnormal phenomena such as smoke or odor occur during the use of the device, please cut off the power immediately, unplug the power cord from the socket, and contact the after-sales service center in time.

Anti-Static Measures

The equipment is a precision electronic device. In order to avoid static electricity from damaging the components, in addition to anti-static measures in the equipment room, you also need to pay attention to the following measures:

- During the installation process (especially when installing the main control board and service board), you must wear anti-static gloves or anti-static wrists.
- When holding the main control board or the service board, try to avoid touching the components or printed circuits.

Grounding Requirements

In order to ensure personal safety and device safety, the device must be grounded.

Power Supply Requirements

The device supports 90 VAC to 264 VAC @ 50/60 Hz power supply. To ensure the stable operation of the device, it is recommended to install UPS for power supply.

Anti-Interference Requirements

- The on-site power supply system must have effective measures to prevent grid interference.
- Do not use the working ground together with the grounding device or lightning protection grounding device of power equipment, and keep the two as far away as possible.
- Keep away from high-power radio transmitters, radar transmitters, and high-frequency and high-current equipment.
- When necessary, electromagnetic shielding can be used for anti-interference.

Environmental Requirements

The device is a system-level monitoring equipment, which is generally placed in the central equipment room of the monitoring system at all levels. The selection of the installation site should comply with the relevant standards of the equipment room construction in the country and region of use.

The device is a standard rack-mounted equipment. Please pay attention to the following information during installation and use:

- Ensure that the temperature in the rack is from 0 °C to 50 °C.
- Ensure that the humidity in the equipment room is between 10% RH and 90% RH (no condensation).
- Ensure that the rack is strong enough to support the weight of the device and its accessories. During the installation, avoid the risk caused by uneven mechanical load.
- Ensure that there is enough installation space for the video and audio cables. The bending radius of a cable should not be less than 5 times the cable outer diameter.
- Keep the horizontal distance between the video wall controller and other devices above 50 cm for sufficient ventilation.

2.2 Open Package and Check Items

Open the device package to verify that all items in the package are intact according to the packing list.

Table 2-1 Packing List

Item	Quantity
Device	1
AC power cord	1
Ground wire	1
Rubber feet	4
Regulatory compliance and safety information manual	1

2.3 Install Modules

The device adopts a plug-and-play modular design, with the main control board managing all service boards. The factory configuration varies by model:

- 2U Model: 1 main control board + 7 blank panels.

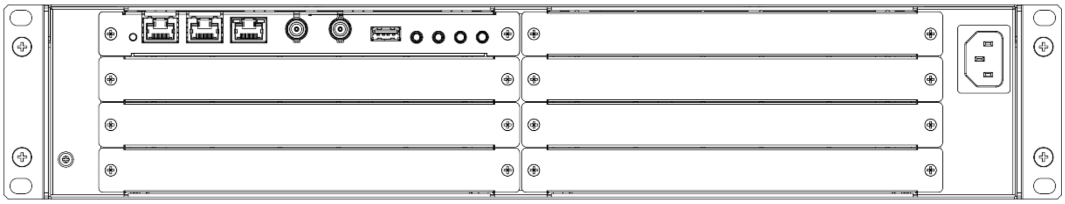

Figure 2-1 Factory Configuration of 2U Device

- 4U Model: 1 main control board + 1 power module + 1 fan module + 13 blank panels.

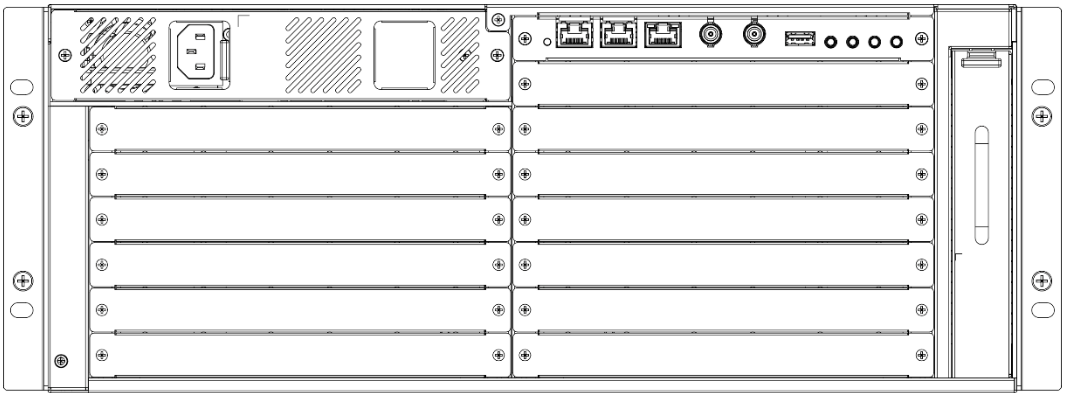

Figure 2-2 Factory Configuration of 4U Device

2.3.1 Install Service Boards

Note

- All service board slots in the 2U device are compatible with both input boards and output boards. In the 4U device, slots S1 to S5 support only input boards, slot S6 is compatible with an input board, a preview board, or an electrical or optical LED controller board, and slots S7 to S12 support only output boards.
- To improve heat dissipation while maintaining low noise levels, when installing a small number of service boards, it is recommended to prioritize slots near the center of the device and adopt a vertically adjacent layout.

Step 1 Use a Phillips screwdriver compatible with M3 screws to remove the fixing screws (1) on both sides of the blank panel, and then pull out the two blank panels from the middle slots.

For normal operation, install at least one input board and one output board. For example, use slots S4 and S5 in a 2U device, or use slots S2 and S9 in a 4U device.

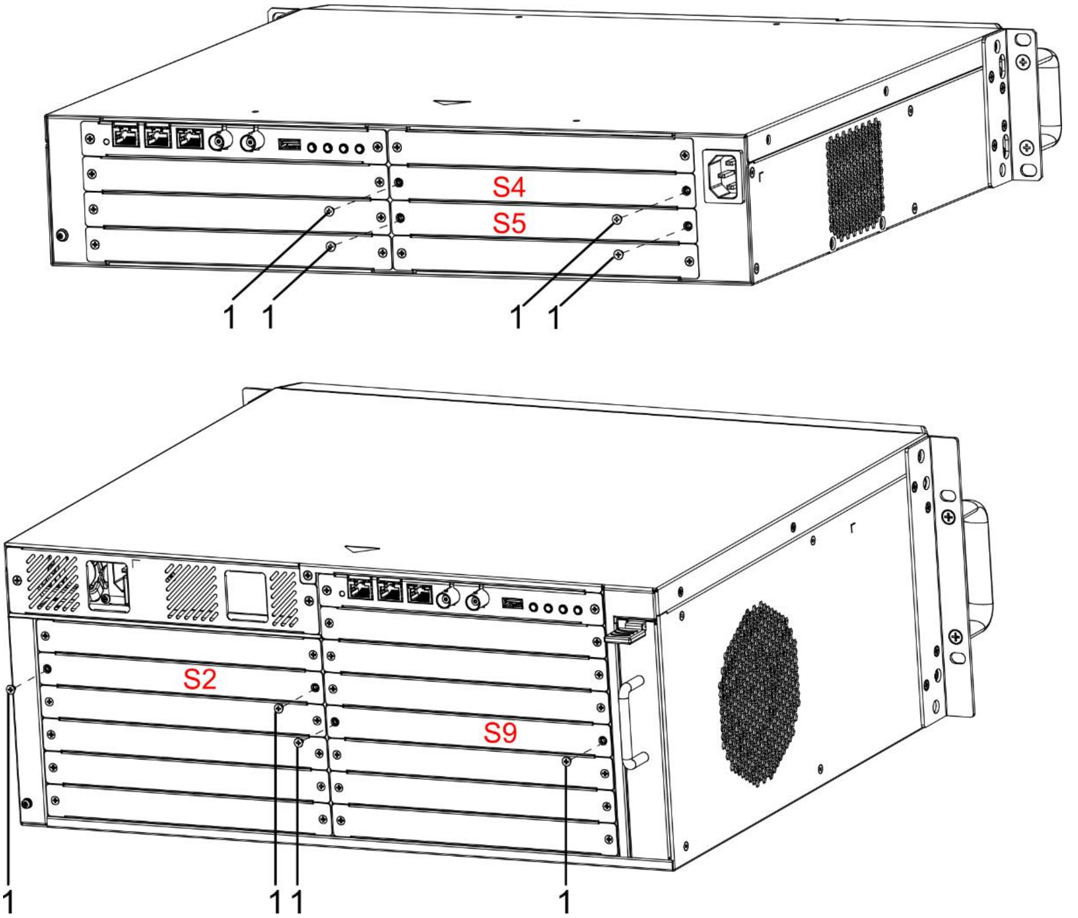

Figure 2-3 Remove Blank Panels

Step 2 Insert the input board (2) and output board (3) into the corresponding slots of the device.

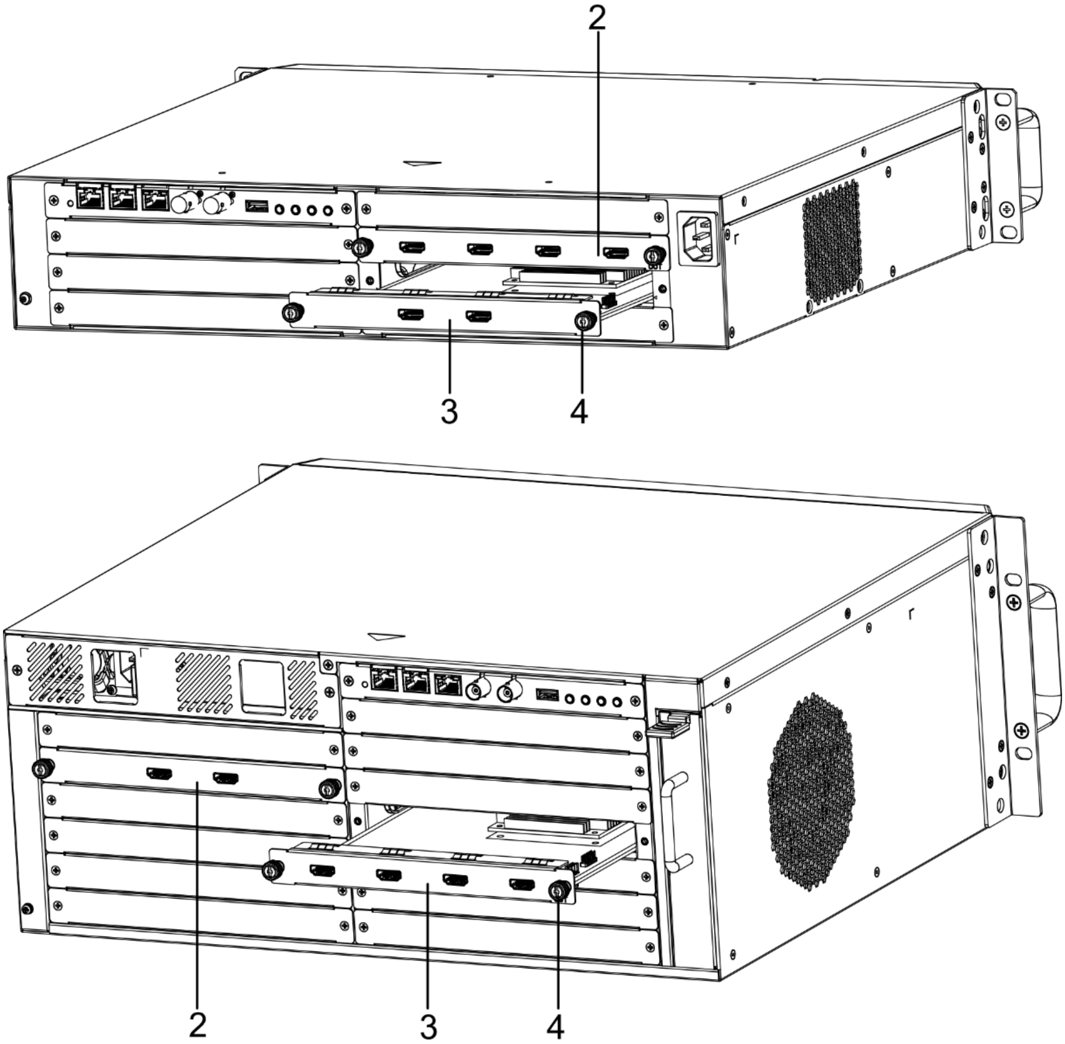

Figure 2-4 Install Two Boards

Step 3 Use the screwdriver to tighten the captive screws (4) provided with the boards clockwise, securing them on both sides of the slots.

Step 4 To install additional service boards, use the screwdriver to remove the fixing screws (1) on both sides of the corresponding blank panel, and then pull out the blank panel.

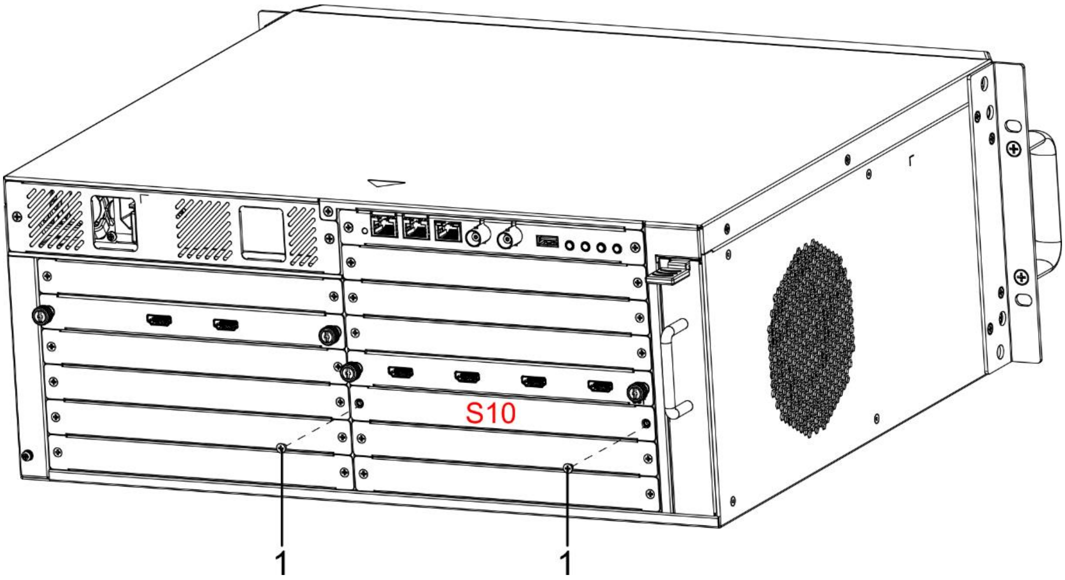

Figure 2-5 Remove Blank Panel

Note

Unused slots must retain blank panels to avoid disrupting the cooling airflow.

Step 5 Insert the service board (5) into the slot along the guide, and then use the screwdriver to tighten the captive screws on both sides of the board clockwise.

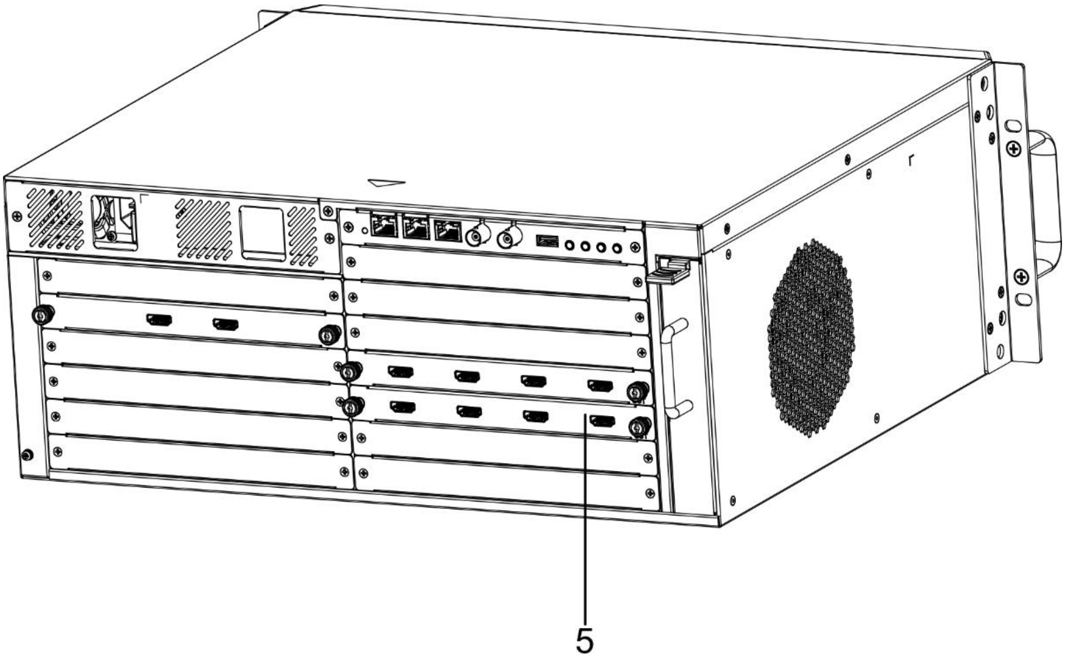

Figure 2-6 Install Three Service Boards

2.3.2 Install Power Module

The 4U device comes with one power module and supports power redundancy by adding an additional power module.

Step 1 Use a Phillips screwdriver compatible with M3 screws to remove the fixing screws (1) on both sides of the power panel.

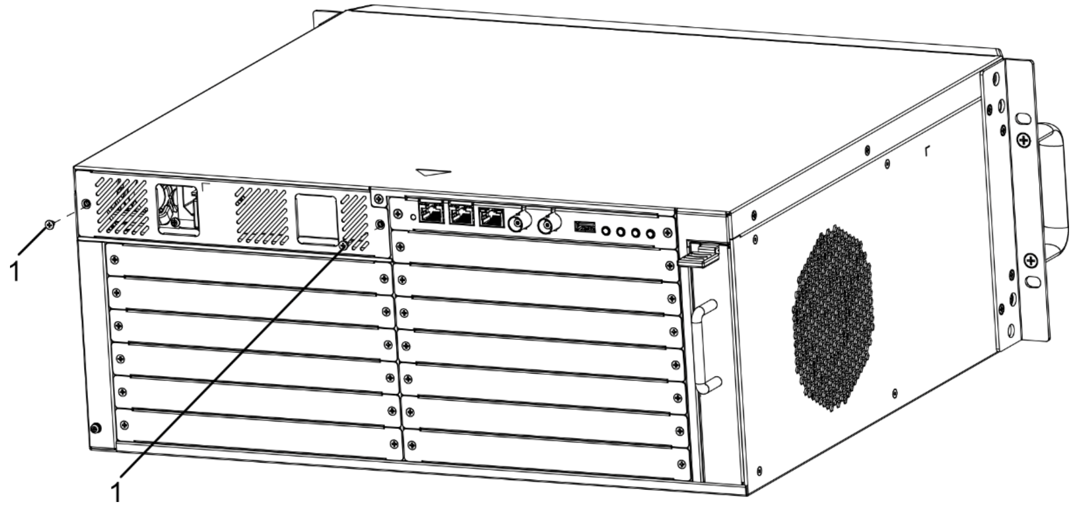

Figure 2-7 Remove Power Panel Fixing Screws

Step 2 Remove the power panel (2).

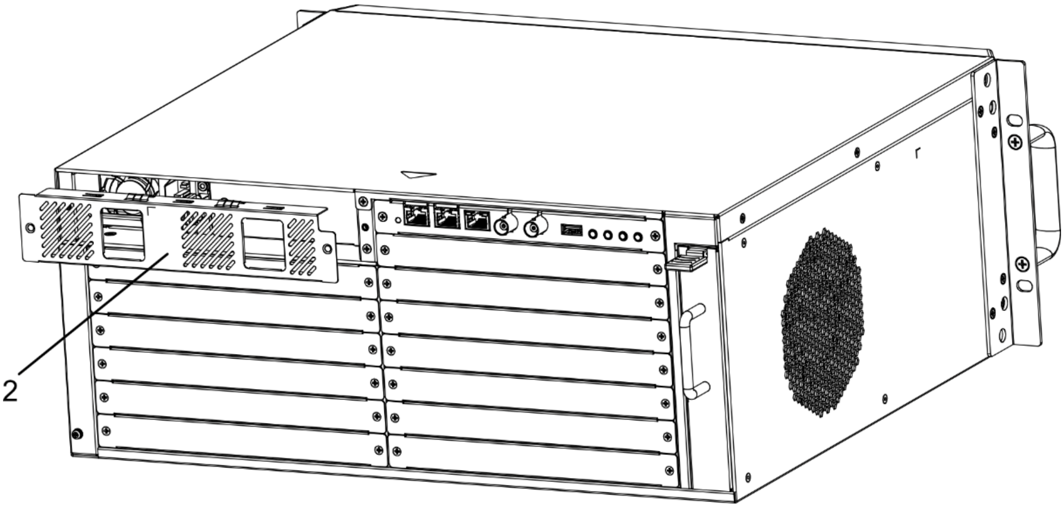

Figure 2-8 Remove Power Panel

Step 3 Use the screwdriver to remove the fixing screws (3) on both sides of the power filler panel (4), and then remove the power filler panel.

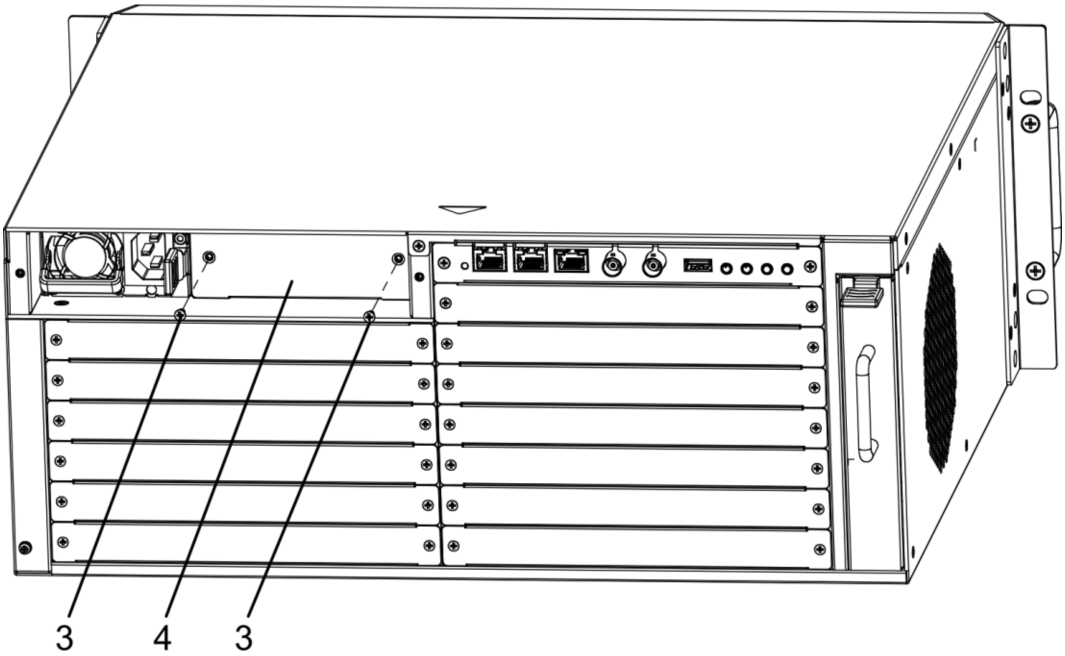

Figure 2-9 Remove Power Filler Panel

Step 4 Insert the power module (5) into the power slot of the device.

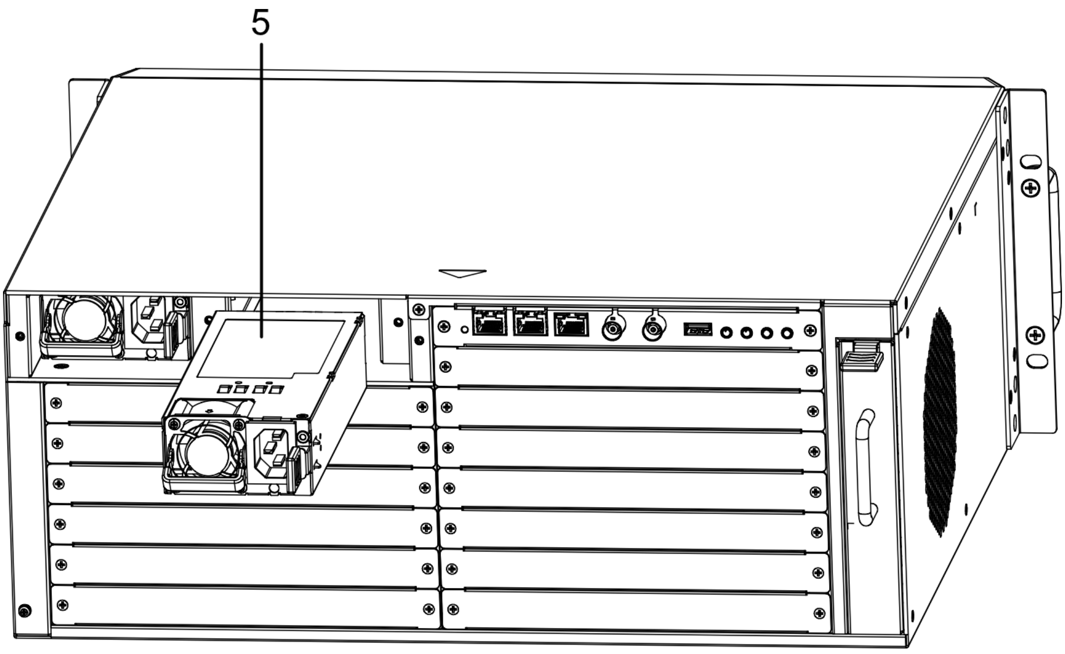

Figure 2-10 Install Power Module

Step 5 Align the power panel with the power slots, and then use the screwdriver to tighten the captive screws on both sides of the panel clockwise, securing it to the device.

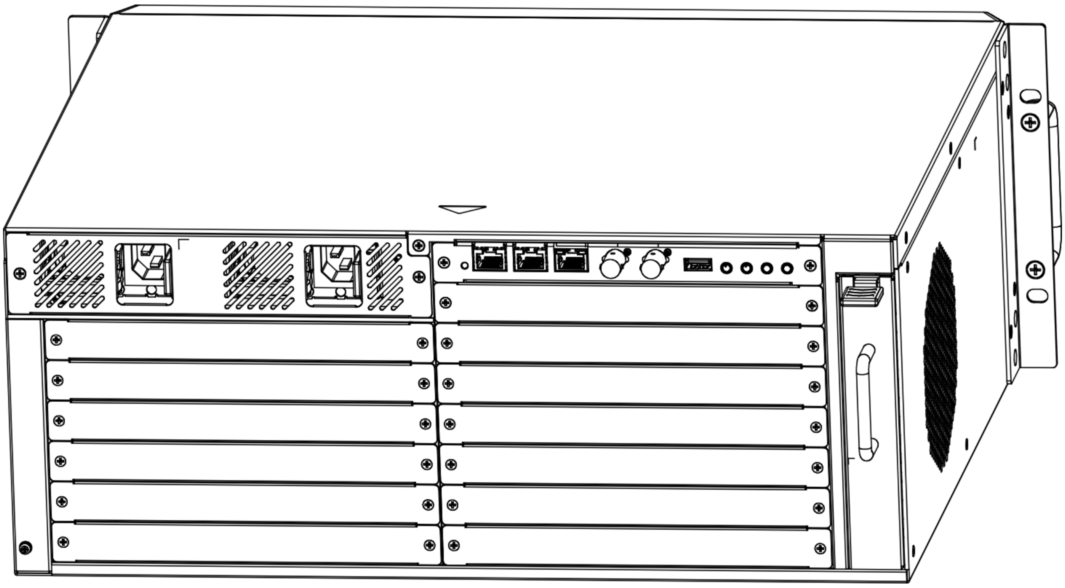

Figure 2-11 Install Power Panel

2.4 Install the Device in the Rack

Both the 2U and 4U devices come with mounting ears pre-installed. Please prepare your own rack. Use the screws (2) provided with the rack to secure the device to the rack posts (1).

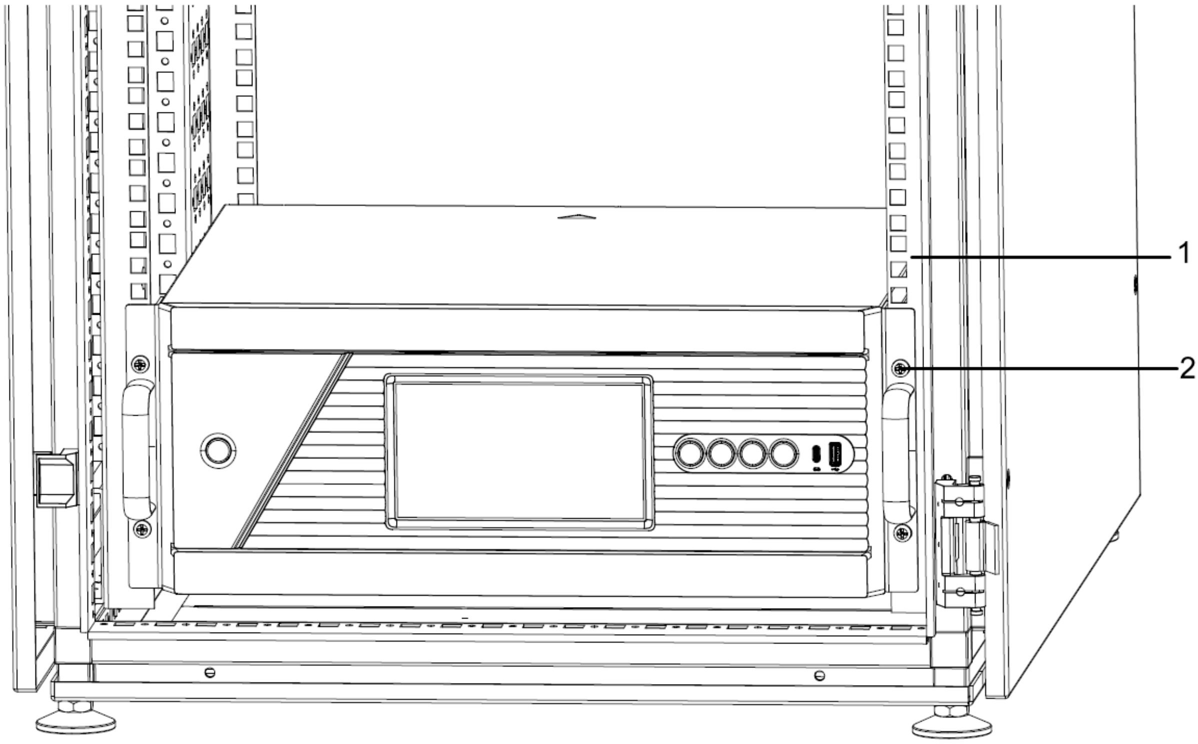

Figure 2-12 Install the 4U Device in the Rack

2.5 Connect the Ground Wire

Connecting the ground wire can release the excessive voltage and current induced by lightning shock. Please select the most suitable connection mode to protect the ground wire according to the installation environment.

Use Grounding Busbar

Step 1 Connect one end of the ground wire (2) to the terminal post of the server room grounding busbar (3).

Step 2 Connect the other end of the ground wire to the equipment grounding terminal (1) and tighten the screw.

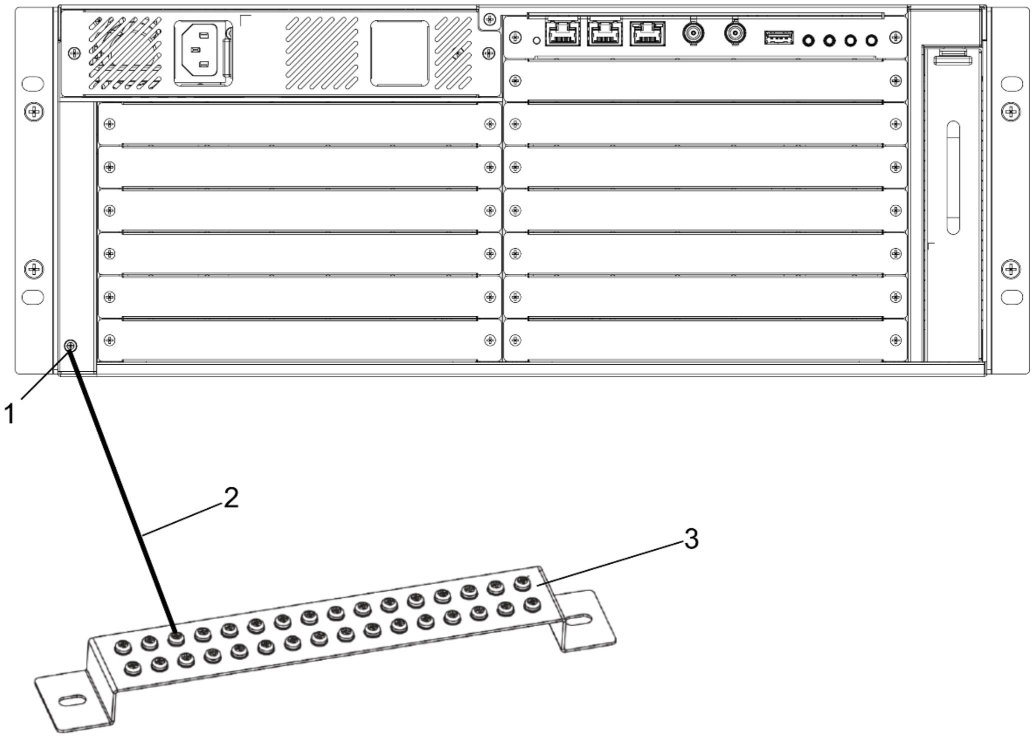

Figure 2-13 Connect the Ground Wire to the Grounding Busbar

Use Grounding Electrode

Step 1 Drive an angle steel or steel pipe (4) with a length ≥ 0.5 m into the ground (3) as a grounding electrode.

Step 2 Weld one end of the ground wire (2) to the grounding electrode and then apply anti-corrosion treatment (e.g., galvanizing or coating) to the welded joint.

Step 3 Connect the other end of the ground wire to the equipment grounding terminal (1).

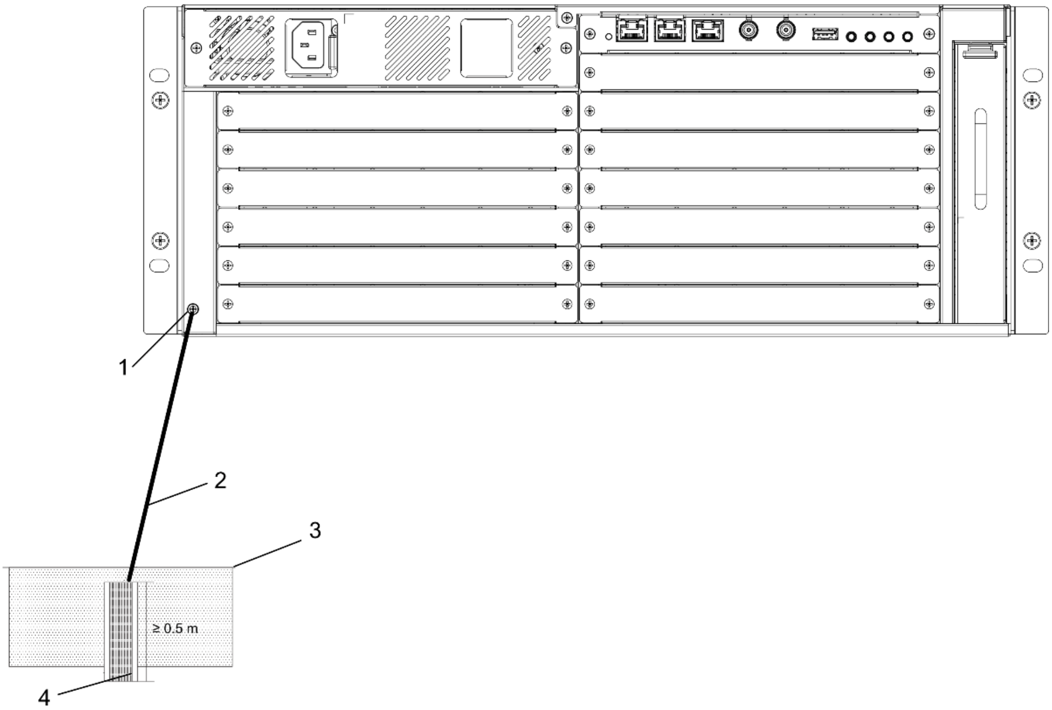

Figure 2-14 Connect the Ground Wire to the Grounding Electrode

2.6 Connect Display and Device

2.6.1 Overview

The device can be connected to both an LCD screen and an LED display simultaneously.

- Connecting an LCD screen: Use the corresponding video cable for connection based on the type of output board installed in the device:
  - HDMI output board: Connect using an HDMI cable.
  - DVI output board: Connect using a DVI cable.
- Connecting an LED display: Select the appropriate connection method based on whether an LED controller board is installed in the device and its type:
  - No LED controller board installed: Connection must be made via an external LED controller. See "2.6.2 Connection via External LED Controller".
  - LED controller board installed: Select the corresponding connection method based on the type of LED controller board and the distance between the display and the device. See "2.6.3 Connection via Electrical LED Controller Board" and "2.6.4 Connection via Optical LED Controller Board".

2.6.2 Connection via External LED Controller

Applicable Scenario

The LED display consists of multiple LED cabinets. Use an external LED controller to connect the LED display when the device is not equipped with an LED controller board. Ensure the number of

Ethernet ports on the external LED controller is not less than the number required for the LED display's daisy-chained links.

Steps

Step 1 Prepare one LED controller.

Step 2 Based on the site layout, divide the LED cabinets into several daisy-chained links. Each link corresponds to one DATA OUT port on the LED controller.

Step 3 Use Ethernet cables (CAT 6 Ethernet cables recommended) to connect the DATA OUT ports of the external LED controller to the Ethernet ports of the first LED cabinet in each respective link. Daisy-chain the subsequent cabinets within each link using Ethernet cables.

Step 4 Use a video cable (HDMI or DVI) to connect the signal output port of the device's output board to the signal input port of the external LED controller.

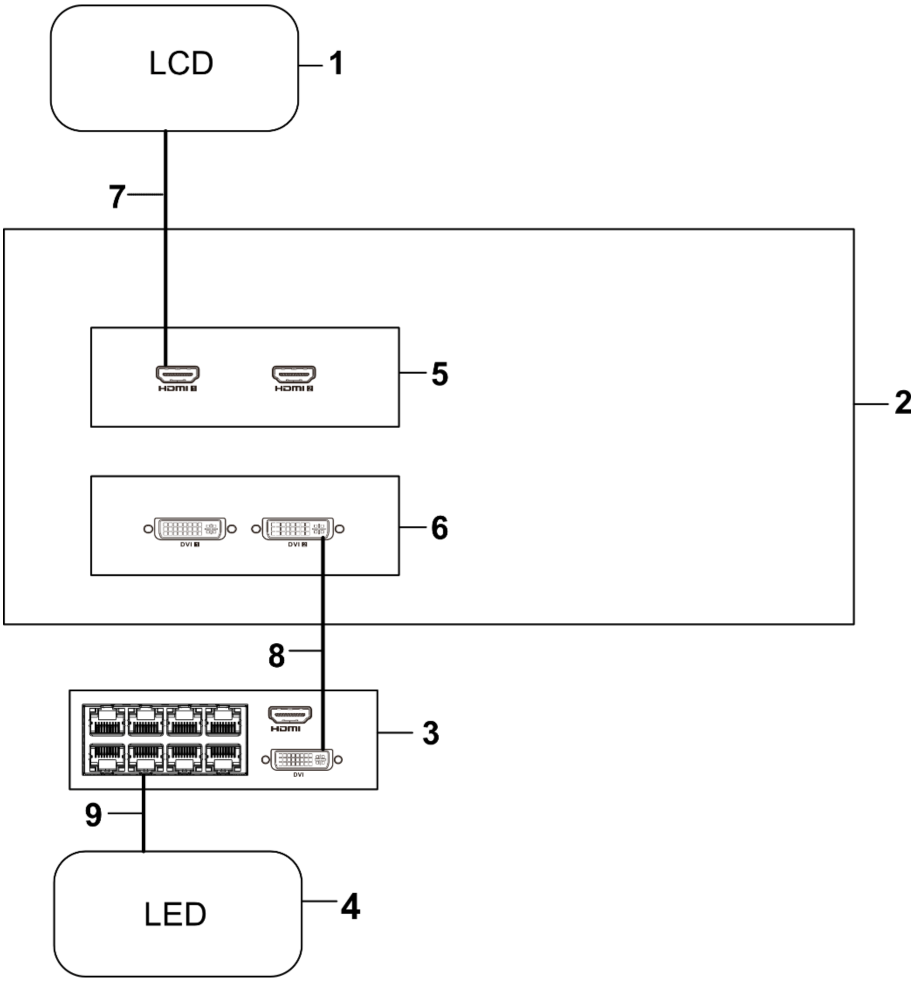

Figure 2-15 Connection via External LED Controller

1. LCD screen	2. Device	3. External LED controller
4. LED display	5. HDMI output board	6. DVI output board
7. HDMI cable	8. DVI cable	9. Eternet cable

2.6.3 Connection via Electrical LED Controller Board

Directly Connect a Local LED Display

Applicable Scenario

Directly connect when the device is equipped with an electrical LED controller board and the LED display is deployed near the device (within the effective transmission distance of an Ethernet cable, typically < 100 meters).

Steps

Step 1 Based on the site layout, divide the LED cabinets into several daisy-chained links. Each link corresponds to one Ethernet port on the electrical LED controller board.

Step 2 Use Ethernet cables (CAT 6 Ethernet cables recommended) to connect the Ethernet ports of the electrical LED controller board to the Ethernet ports of the first LED cabinet in each respective link. Daisy-chain the subsequent cabinets within each link using Ethernet cables.

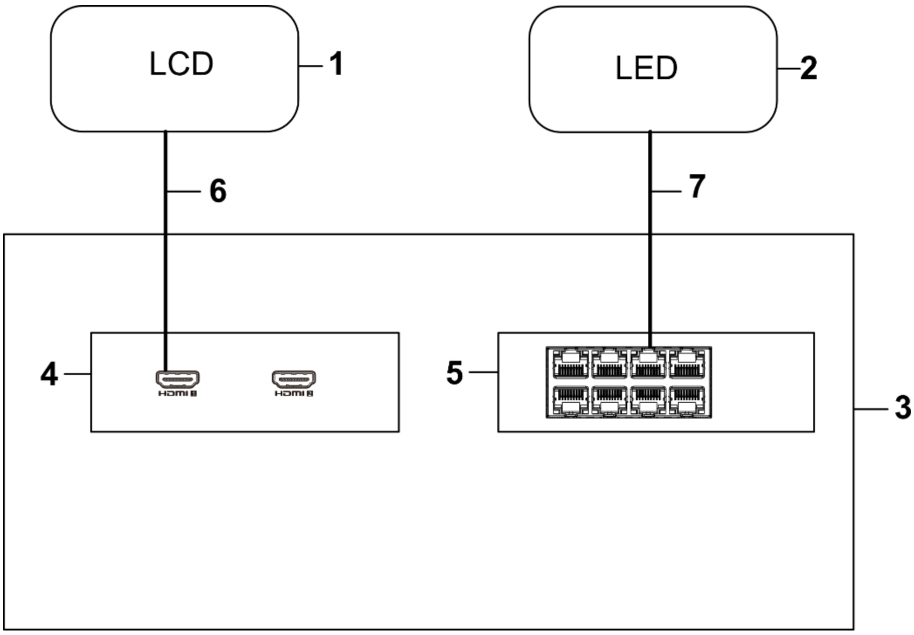

Figure 2-16 Connection via Electrical LED Controller Board

1. LCD screen	2. LED display	3. Device
4. HDMI output board	5. Electrical LED controller board	6. HDMI cable
7. Eternet cable		

Connect a Remote LED Display via Media Converters

Applicable Scenario

Use media converters to extend the connection when the device is equipped with an electrical LED controller board, but the LED display is located far from the device (exceeding the maximum transmission distance of an Ethernet cable, typically > 100 meters). Ensure the number of Ethernet ports on the media converters (transmitter and receiver units) is not less than the number required for the LED display's daisy-chained links.

Steps

Step 1 Based on the site layout, divide the LED cabinets into several daisy-chained links. Each link corresponds to one Ethernet port on the media converter (receiver unit).

Step 2 Use Ethernet cables (CAT 6 Ethernet cables recommended) to connect the Ethernet ports of the media converter (receiver unit) to the Ethernet ports of the first LED cabinet in each respective link. Daisy-chain the subsequent cabinets within each link using Ethernet cables.

Step 3 Use Ethernet cables to connect each Ethernet port of the electrical LED controller board to the Ethernet ports of the media converter (transmitter unit).

Step 4 Use fiber optic patch cords to connect the optical ports of the media converter (transmitter unit) to the optical ports of the media converter (receiver unit).

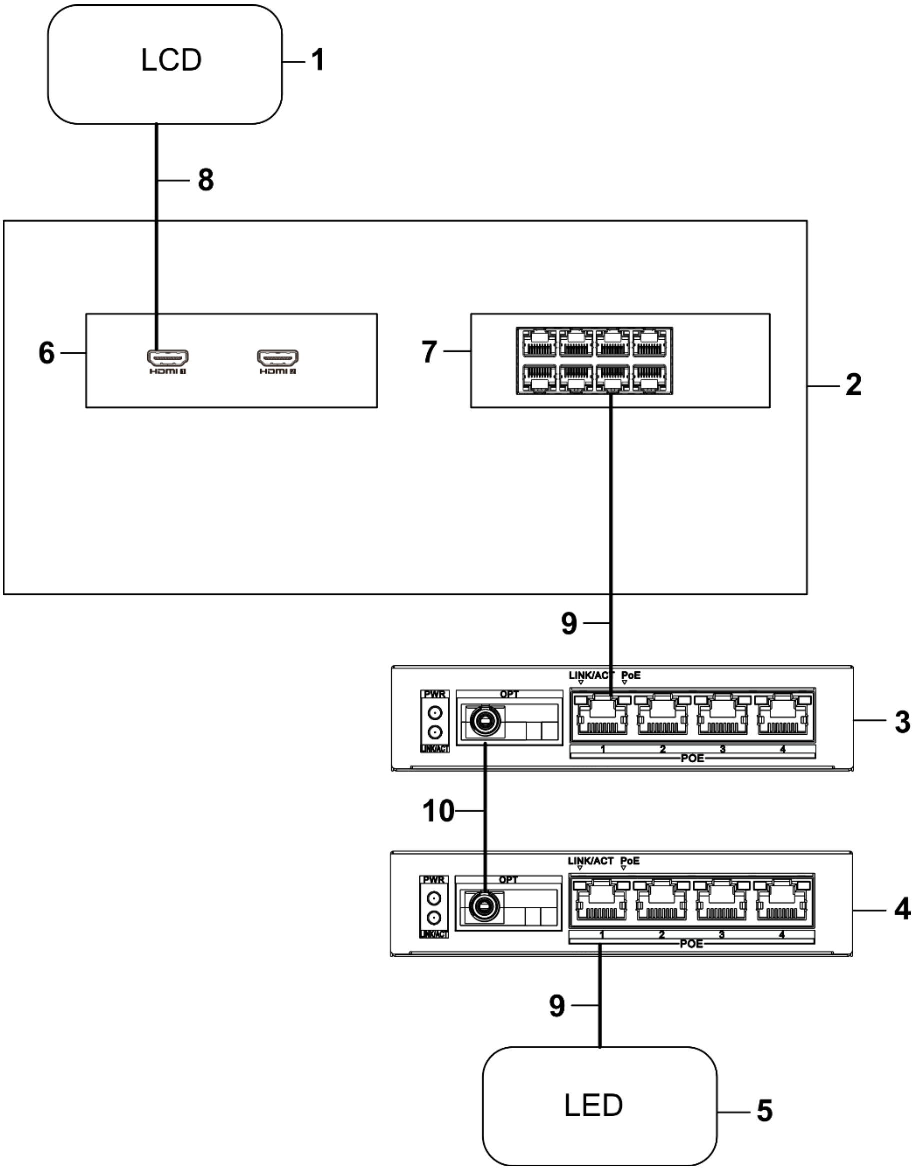

Figure 2-17 Connection via Media Converters

1. LCD screen	2. Device	3. Media converter (transmitter unit)
4. Media converter (receiver unit)	5. LED display	6. HDMI output board
7. Electrical LED controller board	8. HDMI cable	9. Eternet cable
10. Fiber optic patch cord		

Connect a Remote LED Display via Optical Switches

Applicable Scenario

Use optical switches to extend the connection when the device is equipped with an electrical LED controller board, but the LED display is located far from the device (exceeding the maximum transmission distance of an Ethernet cable, typically > 100 meters). Ensure the number of Ethernet ports on the optical switches is not less than the number required for the LED display's daisy-chained links.

Steps

Step 1 Based on the site layout, divide the LED cabinets into several daisy-chained links. Each link corresponds to one Ethernet port on the remote optical switch.

Step 2 Use Ethernet cables (CAT 6 Ethernet cables recommended) to connect the Ethernet ports of the remote optical switch to the Ethernet ports of the first LED cabinet in each respective link. Daisy-chain the subsequent cabinets within each link using Ethernet cables.

Step 3 Use Ethernet cables to connect each Ethernet port of the electrical LED controller board to the Ethernet ports of the local optical switch.

Step 4 Use fiber optic patch cords to connect the optical ports of the local optical switch to the optical ports of the remote optical switch.

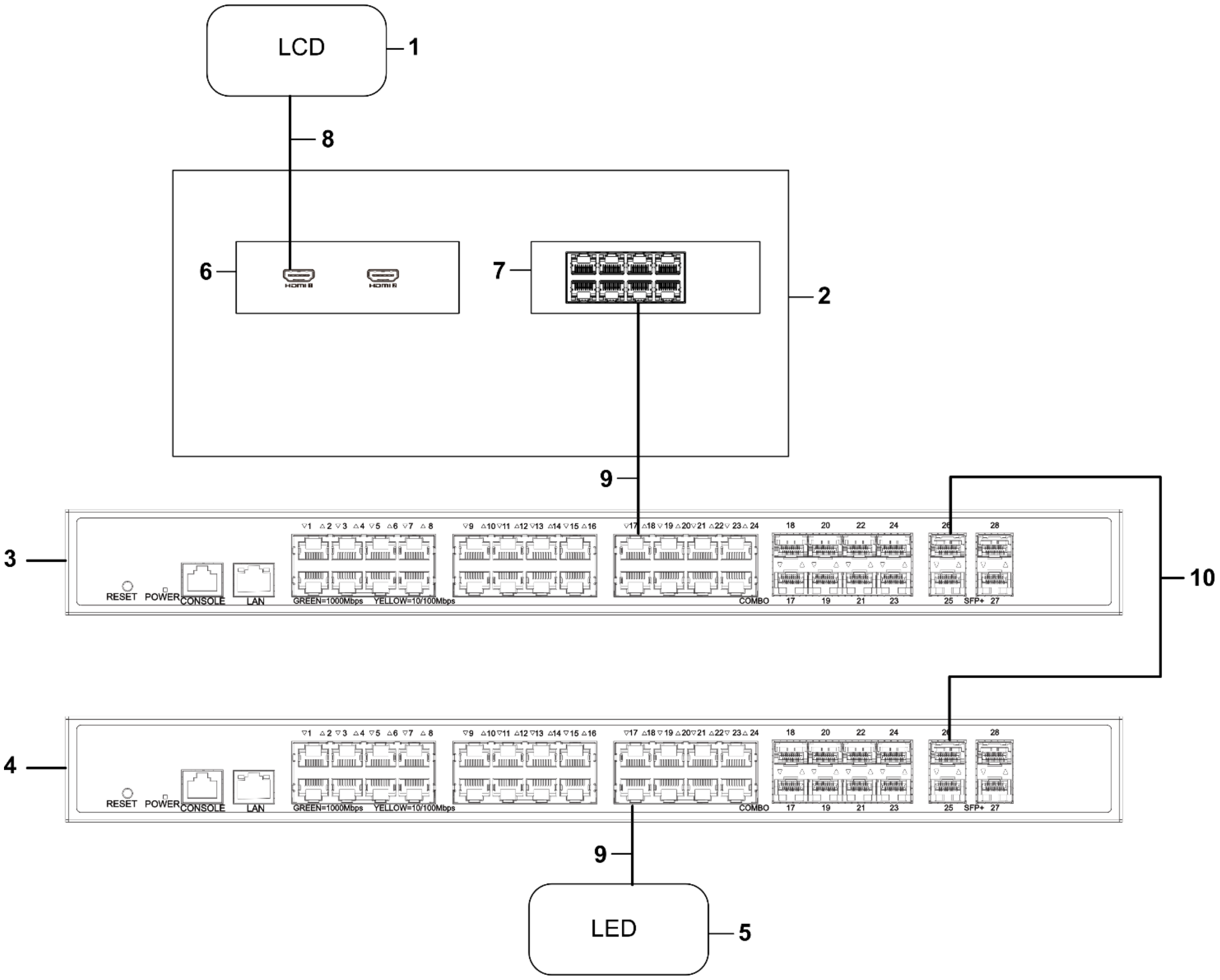

Figure 2-18 Connection via Optical Switches

1. LCD screen	2. Device	3. Local optical switch
4. Remote optical switch	5. LED display	6. HDMI output board
7. Electrical LED controller board	8. HDMI cable	9. Eternet cable
10. Fiber optic patch cord		

2.6.4 Connection via Optical LED Controller Board

Directly Connect a Local LED Display

Applicable Scenario

Directly connect when the device is equipped with an optical LED controller board and the LED display is deployed near the device (within the effective transmission distance of an Ethernet cable, typically < 100 meters).

Important

The optical ports and electrical ports on the optical LED controller board are mutually exclusive options and cannot be used simultaneously.

Steps

Step 1 Based on the site layout, divide the LED cabinets into several daisy-chained links. Each link corresponds to one Ethernet port on the optical LED controller board.

Step 2 Use Ethernet cables (CAT 6 Ethernet cables recommended) to connect the Ethernet ports of the optical LED controller board to the Ethernet ports of the first LED cabinet in each respective link. Daisy-chain the subsequent cabinets within each link using Ethernet cables.

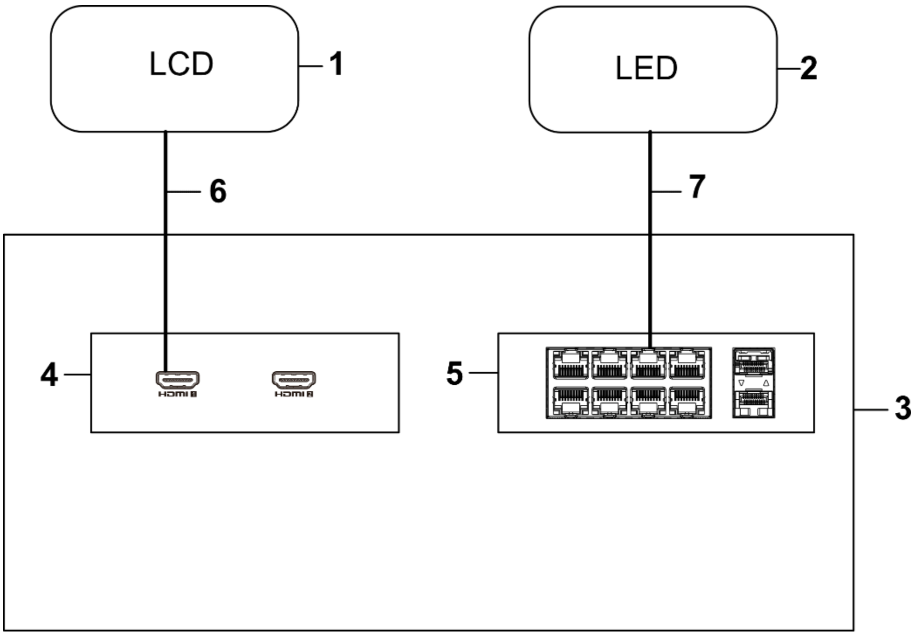

Figure 2-19 Connection via Optical LED Controller Board

1. LCD screen	2. LED display	3. Device
4. HDMI output board	5. Optical LED controller board	6. HDMI cable
7. Eternet cable		

Connect a Remote LED Display via External LED Controller

Applicable Scenario

Use media converters to extend the connection when the device is equipped with an electrical LED controller board, but the LED display is located far from the device (exceeding the maximum transmission distance of an Ethernet cable, typically > 100 meters). Ensure the number of Ethernet ports on the media converters (transmitter and receiver units) is not less than the number required for the LED display's daisy-chained links.

Use fiber optic connection between the device (equipped with an optical LED controller board) and an external LED controller to extend the signal transmission distance when the LED display is far from the device (exceeding the maximum transmission distance of an Ethernet cable, typically > 100 meters). Ensure the number of Ethernet ports on the external LED controller is not less than the number required for the LED display's daisy-chained links.

Important

- The optical ports and electrical ports on the optical LED controller board are mutually exclusive options and cannot be used simultaneously.
- Only external LED controllers with 16 or 24 network ports are supported.

Steps

Step 1 Prepare one LED controller equipped with optical ports.

Step 2 Based on the site layout, divide the LED cabinets into several daisy-chained links. Each link corresponds to one DATA OUT port on the LED controller.

Step 3 Use Ethernet cables (CAT 6 Ethernet cables recommended) to connect the DATA OUT ports of the external LED controller to the Ethernet ports of the first LED cabinet in each respective link. Daisy-chain the subsequent cabinets within each link using Ethernet cables.

Step 4 Use a fiber optic patch cord to connect the optical port of the device's Optical Port LED Controller Board to the optical port of the external LED controller.

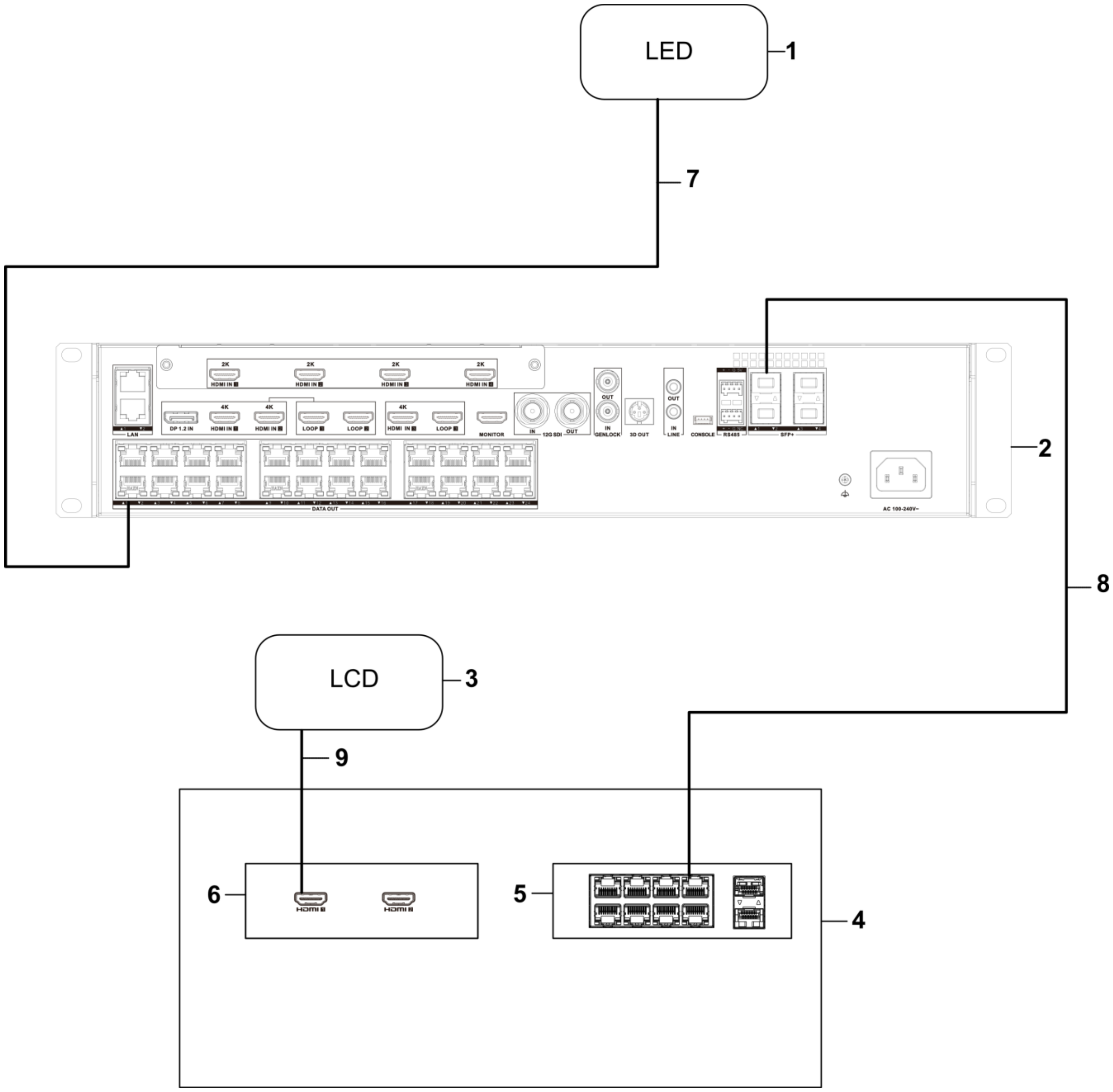

Figure 2-20 Remote LED Display Connection

1. LED display	2. External LED controller	3. LCD screen
4. Device	5. Optical LED controller board	6. HDMI output board
7. Eternet cable	8. Fiber optic patch cord	9. HDMI cable

2.7 Connect the Power Cord

Use a power cord to connect the power supply socket of the device to the power supply in the equipment room. After the power cord is connected, the device is powered on.

Chapter 3 Configuration

3.1 Activate Device

You should activate the device before using the device for the first time. You can use the SADP client or the device web page to activate the device. When activating the device, obey the following requirements to set the password:

- To improve system security, it is highly recommended to change password regularly. In order to protect your privacy and corporate data, and avoid network security issues, it is recommended to set strong password that meets security requirements.
- Password should contain 8 to 16 characters and at least 2 of the following types: digits, lowercase letters, uppercase letters, and special characters.
- Password cannot contain user name, 123, admin (case insensitive), 4 or more continuously ascending or descending digits, or 4 or more consecutive repeated characters.

Activate Devices via HiTools Delivery Client

Step 1 Connect all devices and the computer to the same LAN, ensuring they are on the same IP subnet.

Step 2 Install and launch the HiTools Delivery client on the computer.

Step 3 Navigate to **Device Management** > **Current Subnet**, and click **Refresh**.

Step 4 Select the inactive devices, set the activation password, confirm the password, and click **Activation**.

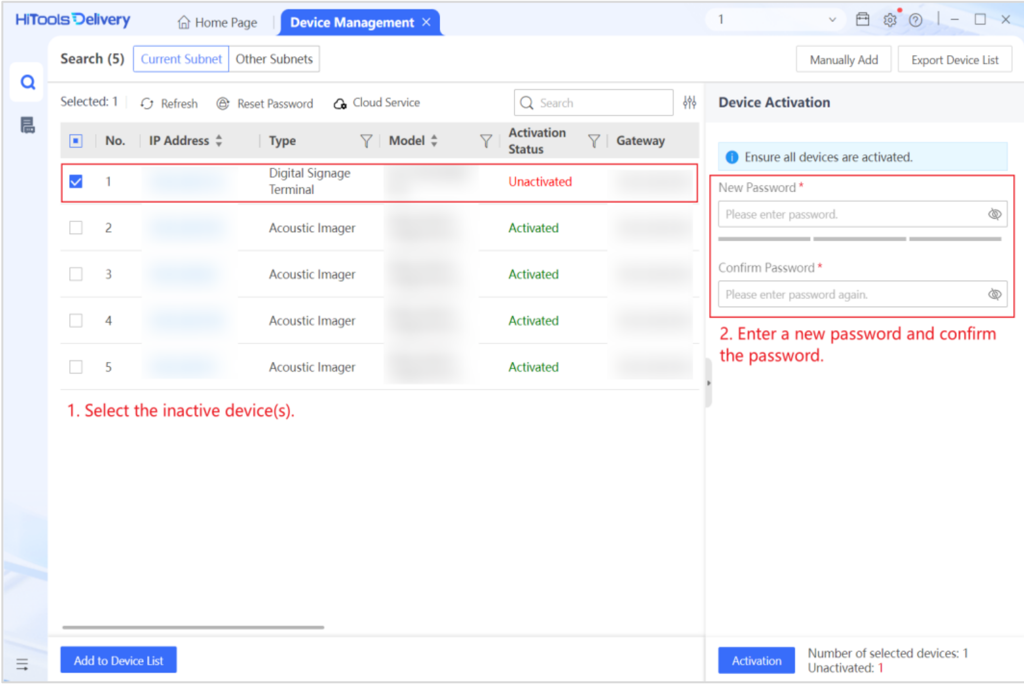

Figure 3-1 Batch Activate Devices

Step 5 Edit device IP addresses in batch:

1) Check multiple activated devices.
2) Choose one of the following methods to set IP addresses:
   - Manual assignment: Set the start IP address, port No., IPv4 subnet mask, IPv4 gateway, etc., and the selected devices will be automatically assigned to increasing IP addresses.
   - Dynamic acquisition: Check **Enable DHCP** to assign dynamic IP addresses.
3) Enter the admin password and click **OK**.

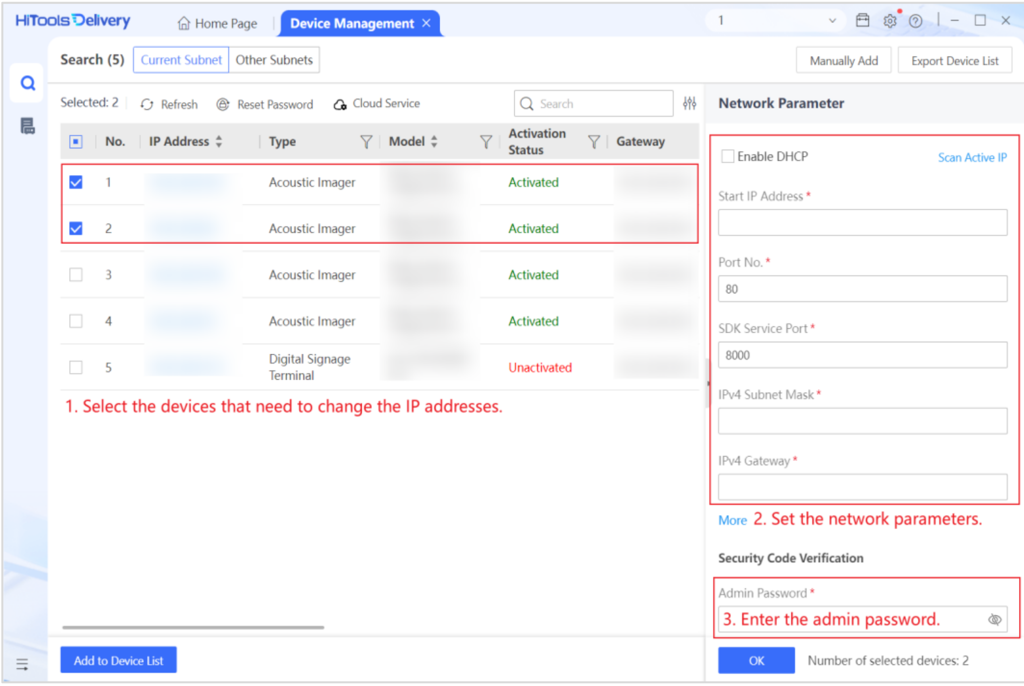

Figure 3-2 Batch Edit Device IP Addresses

Activate the Device via SADP Client

Step 1 Connect the device and computer to the same LAN. Make sure the device and computer in the same network segment.

Step 2 Download and install the SADP client on the computer.

Step 3 Open the SADP client.

Step 4 Select the device that is not activated, enter the activation password and confirm it, and click **Activate**.

If the device cannot be found, you can restart the SADP client.

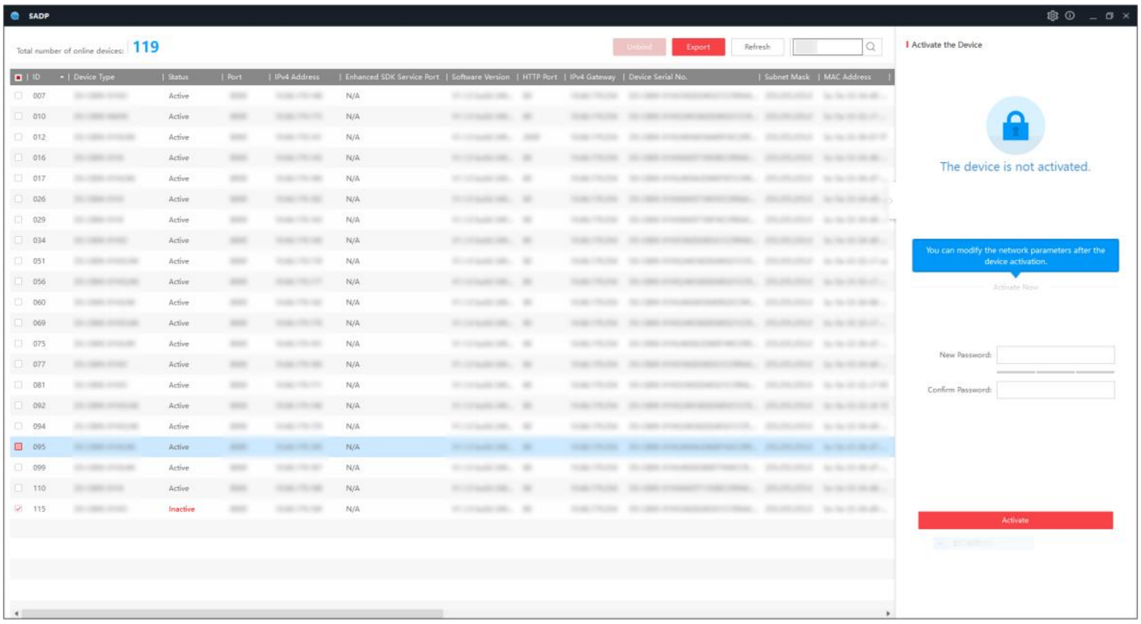

Figure 3-3 Activate the Device via SADP Client

Step 5 View the device IP address in the SADP client and enter the device IP address in the computer browser.

Activate the Device via Web Browser

Step 1 Use a network cable to connect a computer to the device.

Step 2 Set the computer IP address to any IP address in the range of 192.0.0.2 to 192.0.0.253 (excluding 192.0.0.64) and set the computer gateway address to 192.0.0.1.

By default, the device IP address is 192.0.0.64 and the gateway address is 192.0.0.1.

Step 3 Enter 192.0.0.64 in the computer browser to enter the device activation page.

Step 4 Set the activation password, and then click **Activate**.

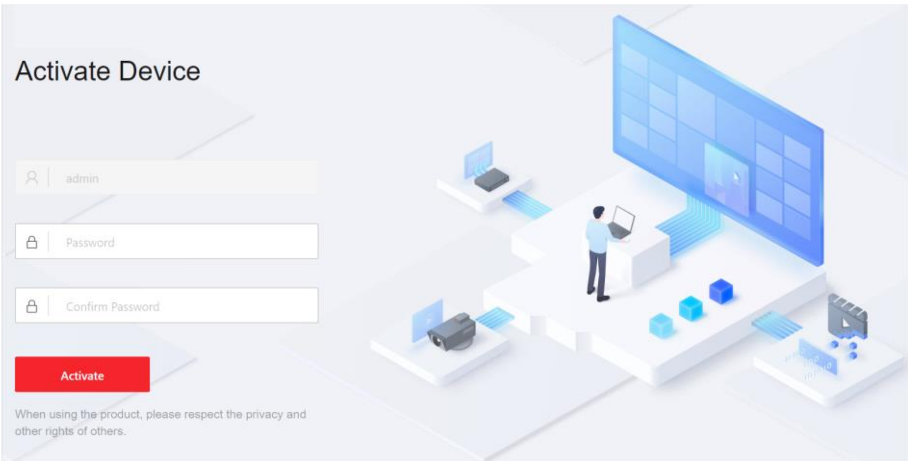

Figure 3-4 Activate the Device via Browser

3.2 More Configuration

Scan the QR code below to view the user manual to configure the device.

Note

Obtaining the manual requires network data traffic. It is recommended to be performed in a Wi-Fi environment.

Figure 3-5 User Manual

See Far, Go Further

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